

Assignment-2

1. A transmission line has a span of 200 m between level supports. The conductor has cross-sectional area of 1.29 cm^2 , weight $1,170 \text{ kg/m}$ and has a breaking stress of $4,218 \text{ kg/cm}^2$. Calculate sag for a safety factor of 5 allowing a wind pressure of 122 kg/m^2 of projected area. What is vertical sag?

2. An overhead line having a span of 300 m has ACSR conductors of 1.95 cm diameter. The allowable tension is 35,600 N. Find

- Sag in still air condition with no ice covering.
- Vertical sag when there is an ice coating of 0.8 cm and a horizontal wind pressure of 400 N/m^2 of projected area. Ice density = $8,900 \text{ N/m}^3$.
- What ~~will~~^{should} be the height of the lowest cross-arm to give a minimum ground clearance of 7.8 m under bad weather conditions?

Line is carried by insulator string 1.6 m long.
Weight of conductor = 0.85 kg/m .

3. for a string insulator with 4 discs, the capacitance of the disc is 10 times the capacitance between the pin and earth. Calculate the voltage across each disc when used on 66 kV line. Also calculate the string efficiency.

6. A string of six insulator units has self capacitances equal to 10 times the pin to earth capacitances.

Find

(i) the voltage distribution from top to bottom insulator as a percentage of the total voltage.

(ii) String efficiency.

5. List the basic tests that are carried out on insulators.